

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$| -5x + 13 | = 73$

What is the sum of the solutions to the given equation?

Answer

- A. $-\frac{146}{5}$
- B. -12
- C. 0
- D. $\frac{26}{5}$

Correct Answer: D

Rationale

Choice D is correct. By the definition of absolute value, if $| -5x + 13 | = 73$, then $-5x + 13 = 73$ or $-5x + 13 = -73$. Subtracting 13 from both sides of the equation $-5x + 13 = 73$ yields $-5x = 60$. Dividing both sides of this equation by -5 yields $x = -12$. Subtracting 13 from both sides of the equation $-5x + 13 = -73$ yields $-5x = -86$. Dividing both sides of this equation by -5 yields $x = \frac{86}{5}$. Therefore, the solutions to the given equation are -12 and $\frac{86}{5}$, and it follows that the sum of the solutions to the given equation is $-12 + \frac{86}{5}$, or $\frac{26}{5}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is a solution, not the sum of the solutions, to the given equation.

Choice C is incorrect and may result from conceptual or calculation errors.